

International Institute of Information Technology, Bhubaneswar

Department of Computer Science and Engineering

Object Oriented Programming Laboratory**Lab 3 – C++ Programming: Dynamic Memory Allocation****Date:** 11.03.2026**Semester:** B.Tech 3rd Semester**Section:** CSE B2**Instructions**

- Write all programs in C++.
- Use the new operator for dynamic memory allocation wherever required.
- Release all dynamically allocated memory using the appropriate delete or delete[] operator.
- Display the output in a clear and readable format.
- Ensure that dynamically allocated memory is properly deallocated before the program terminates.

1. Dynamic Integer Allocation

Write a C++ program to dynamically allocate memory for a single integer using the new operator. Read an integer from the user, store it in the dynamically allocated memory, display its value, and release the allocated memory using the delete operator. *Hint: Use an integer pointer to store the address returned by new.*

2. Dynamic Array of Integers

Write a C++ program to dynamically allocate an array of n integers using the new operator. Read the elements from the user and display them. Finally, release the allocated memory using the appropriate form of the delete operator. *Hint: Use new int[n] for allocation and delete[] for deallocation.*

3. Find the Largest Element

Write a C++ program that dynamically allocates an array of n integers. Accept the elements and determine the largest element in the array. After displaying the largest element, properly deallocate the allocated memory. *Hint: Use a pointer to access the dynamically allocated array.*

4. Dynamic Array and Average

Write a C++ program that dynamically allocates memory for n floating-point numbers. Accept the numbers, calculate their sum and average, and display the results. Ensure that the dynamically allocated memory is released after use. *Hint: Use new float[n] for allocation and delete[] for deallocation.*

5. Dynamic Object Creation

Create a class named Student with the following data members:

- Roll Number
- Name
- Marks

Create an object of the class dynamically using the `new` operator. Write member functions to:

1. Accept student details.
2. Display student details.

Finally, release the dynamically allocated object using `delete`. ***Hint:** For a dynamically allocated object, member functions can be accessed using the `->` operator.*

6. Array of Dynamic Objects

Create a class named Employee containing:

- Employee ID
- Employee Name
- Salary

Dynamically allocate memory for n Employee objects using the `new` operator. Accept and display the details of all employees. Finally, release the allocated memory correctly. ***Hint:** Use `new Employee[n]` for allocation and `delete[]` for deallocation.*

7. Dynamic Matrix

Write a C++ program to dynamically allocate memory for a two-dimensional matrix of size $m \times n$. Accept the matrix elements and display the matrix. Finally, deallocate all dynamically allocated memory properly. ***Hint:** You may use a pointer-to-pointer approach (`int **matrix;`). Allocate memory for the rows first and then allocate memory for each row.*

8. Dynamic Student Marks System

Create a class named Student that stores:

- Roll Number
- Name
- Number of Subjects
- Dynamically allocated array of marks

The number of subjects should be entered by the user at runtime. Implement member functions to:

1. Accept student details.
2. Dynamically allocate memory for the marks.
3. Accept marks for all subjects.
4. Calculate total and average marks.
5. Display the complete result.
6. Properly release the dynamically allocated marks.

Hint: The marks array should be dynamically allocated inside the class using the `new` operator.

9. Dynamic Shopping Cart

Create a class named `Product` containing:

- Product ID
- Product Name
- Price
- Quantity

Create a shopping cart that can store n products, where n is entered by the user at runtime. Dynamically allocate memory for the products and implement member functions to:

1. Accept product details.
2. Display all products.
3. Calculate the total cost of the shopping cart.
4. Display the total amount.
5. Release the dynamically allocated memory.

Hint: Use a dynamically allocated array of `Product` objects.

10. Dynamic Employee Salary Analysis

Create a class named `Employee` containing:

- Employee ID
- Employee Name
- Basic Salary
- Dynamically allocated array for monthly earnings

The number of months should be entered by the user at runtime. Write a program that:

1. Dynamically allocates memory for the required number of months.
2. Accepts the employee's monthly earnings.
3. Calculates the total earnings.
4. Calculates the average monthly earning.
5. Finds the month with the highest earning.
6. Displays the complete analysis.
7. Properly deallocates all dynamically allocated memory.

Hint: Use `new` for runtime memory allocation and ensure that every dynamically allocated memory block has a corresponding `delete` or `delete[]`.